Thin Solid Films, *169* (1989) 229–233
PREPARATION AND CHARACTERIZATION

THE ANNEALING BEHAVIOUR OF ELECTROLESS NONCRYSTALLINE NICKEL PHOSPHORUS FILMS*

S. V. S. TYAGI, S. K. BARTHWAL AND V. K. TONDON
Department of Physics, University of Roorkee, Roorkee 247667 (India)
S. RAY
Department of Metallurgical Engineering, University of Roorkee, Roorkee 247667 (India)
(Received December 7, 1987)

The annealing behaviour of the resistivity and magnetization of electroless amorphous Ni–P films has been studied and corroborated by electron microscopy. The changes observed may be attributed to atomic relaxation, the precipitation of crystalline nickel from the amorphous matrix, the crystallization of the film and some internal structural changes.

1. INTRODUCTION

The study of the crystallization characteristics of electrodeposited amorphous Ni–P alloys was recently taken up by Bakonyi and coworkers[1–3]. Few earlier studies exist[4], and even for the case of electrodeposited Ni–P, there appears to be no consensus as to the steps involved in crystallization, or the dependence of these steps on composition or temperature for electrodeposited Ni–P. For electroless Ni–P, no systematic work is known to exist at all.

This paper presents a study of the crystallization behaviour of electroless Ni–P through the annealing responses of both the resistivity and the magnetic moment of the noncrystalline phase in the temperature range 300–800 K. Hot-stage electron microscopy has been used to integrate all the information so that a clear picture can be presented.

From the structural perspective, the electroless Ni–P system may be divided into four categories: (i) polycrystalline films having less than 11 at.% P, (ii) microcrystalline films with 11–14 at.% P, (iii) films with 14–18 at.% P having both microcrystalline and amorphous regions; (iv) amorphous films that contain more than 18 at.% P.

Both in the polycrystalline and microcrystalline films, the as-deposited state of Ni–P is a supersaturated solution of phosphorus dissolved in nickel. The grain size decreases with increasing amounts of phosphorus[4–6]. The nature of the as-deposited state for the electroless films containing more than 14 at.% P has been the subject of

* Paper presented at the 7th International Conference on Thin Films, New Delhi, India, December 7–11, 1987.

0040-6090/89/$3.50
© Elsevier Sequoia/Printed in The Netherlands

our earlier work[7–9]. On heating, Ni–P films, irrespective of their initial phosphorus content, transform into an equilibrium mixture of nickel and Ni_3P, with the amount of nickel relative to Ni_3P decreasing with the increase in the amount of phosphorus. The mechanism of transformation for the polycrystalline and microcrystalline films is simply the nucleation and growth of the Ni_3P, and the grain size coarsening of the nickel phase. However, for films containing more than 14 at.% P, the complexity of the transformation increases and several additional steps can be identified.

Here, we report the results of our annealing studies of Ni–P films with a phosphorus content of more than 18 at.%, in which the as-deposited state is known to consist of an amorphous region alone[9].

2. EXPERIMENTAL DETAILS

The samples were prepared using the electroless deposition technique. An alkaline sulfate bath, details of which are described elsewhere[8,9], was used. For the chemical analysis we employed a wet chemical method and a magnetic moment method[10].

Electron microscopy was carried out with a Phillips electron microscope model EM 400 T/ST with a heating holder attachment. The resistance was measured using a standard four-probe method, within an accuracy of about 1%. However, the systematic error in resistivity of a given sample is estimated to be about 10%, due to its geometry. The magnetic moment measurements were done on a vibrating sample magnetometer model 155 EG&G PARC, with a temperature oven and a temperature controller attached.

3. RESULTS AND DISCUSSION

The diffraction pattern of the as-deposited samples containing 18–21 at.% phosphorus shows only one diffuse ring and a very weak outer ring. This is characteristic of amorphous Ni–P deposits and is distinctly different from patterns obtained from microcrystalline samples. A further confirmation of the amorphous nature of the deposits has been obtained from the annealing studies reported elsewhere[8,9].

The temperature dependence of the relative resistivity of an amorphous Ni–P alloy containing 18.8 at.% P is shown in Fig. 1. The resistivity increases up to about 420 K, and then shows a minor decrease (region a) over a broad temperature range extending up to about 540 K. Further, a sharp decrease in resistivity to about 75% of its original value (region b) is observed between temperatures of 540 and 580 K. The resistivity shows a further drop of 5%–10%, but at a much slower rate (region c), in the temperature range 580–740 K. When cooled, the room-temperature resistivity can be seen to be markedly lower as compared with that of the as-deposited state. Further thermal cycling of the sample produces reversible behaviour, as expected. These changes in resistivity have been observed reproducibly in a large number of films of similar composition.

The annealing behaviour of magnetization is shown in Fig. 2. The films with 19.7 and 20.3 at.% P show two peaks, but the film with 23.5 at.% P shows only one small peak.

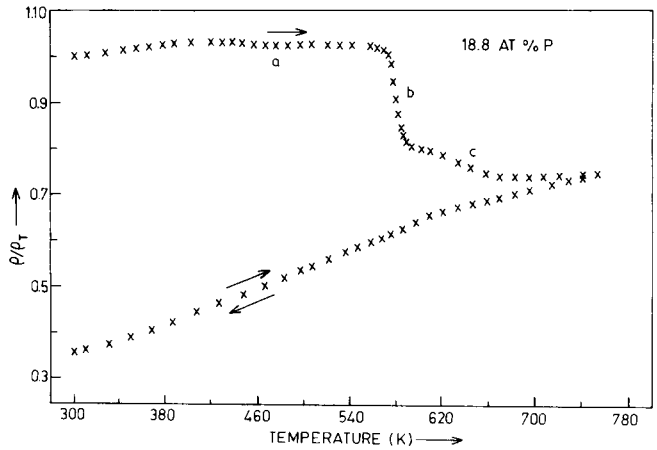


Fig. 1. Annealing behaviour of the relative resistivity of amorphous Ni–P containing 18.8 at.% P.

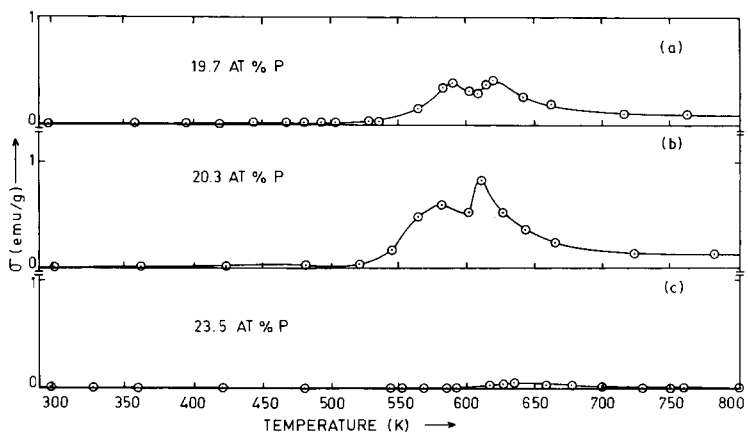


Fig. 2. Annealing behaviour of the magnetization of amorphous Ni–P with the following phosphorus contents: (a) 19.7 at.% P; (b) 20.3 at.% P; (c) 23.5 at.% P.

Figure 3 shows the diffraction pattern of films heated to 423 K, 573 K and 643 K for a period of 2 min. The sample heated to 423 K still shows diffuse rings; on heating to 573 K crystalline nickel can be observed in the amorphous matrix; on heating to 643 K, it can be seen that the process of crystallization is complete and that the amorphous phase is not present. In addition to crystalline nickel, the spot pattern of Ni_3P is observed.

If the observations on the changes in resistivity, magnetization and diffraction pattern on heating are combined, the underlying transformations can be understood to some extent. The change in resistivity, corresponding to region (a) in Fig. 1, at about 420 K does not reveal any accompanying change in diffraction pattern or magnetization. It can be inferred that the small change in resistivity may be due to atomic relaxation which is normally observed in amorphous films but not detected either by diffraction or by a change in the magnetization response[11].

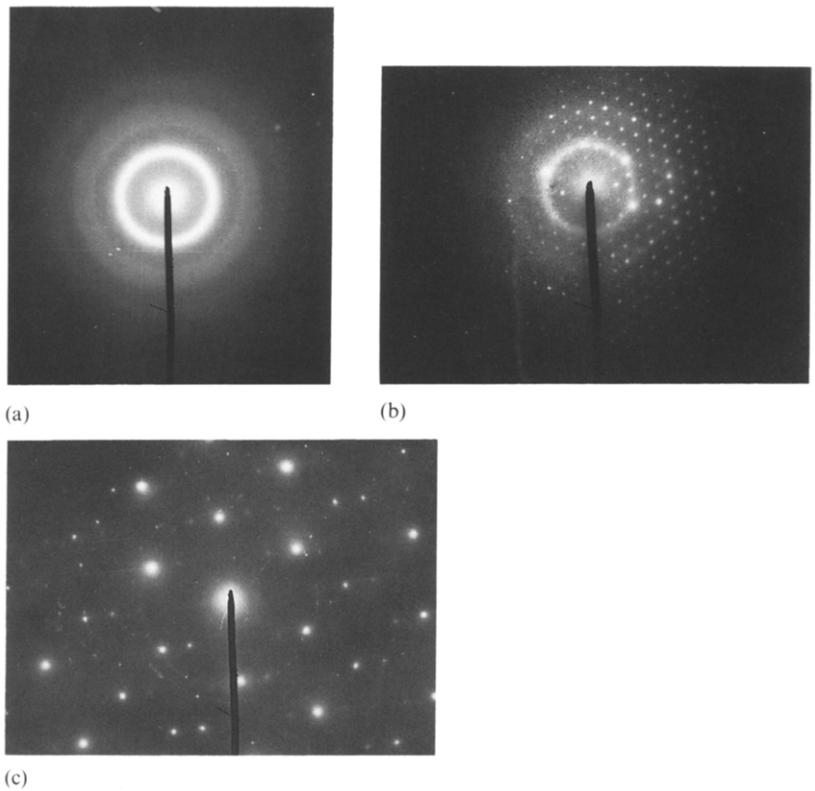


Fig. 3. Diffraction pattern of the amorphous Ni–P sample heated for 2 min to the following temperatures: (a) 423 K; (b) 573 K; (c) 643 K.

The start of the first peak of magnetization around 540 K is attributed to precipitation of crystalline nickel from the amorphous matrix. This is evident from the co-existence of diffraction spots of crystalline nickel with the diffuse rings from the amorphous phase, as shown in Fig. 3(b). No corresponding change in resistivity is observed however. This may be due to the discontinuous and dispersed nature of nickel crystallites. The extent of this reaction decreases with increasing phosphorus in the film, as can be seen from Fig. 2(c).

The sharp drop in resistivity in region (b), accompanied by the disappearance of rings corresponding to the amorphous phase, as can be seen from Fig. 3(c), indicates the crystallization of the matrix. This is also evident from the second peak in the magnetization, corresponding to the formation of an equilibrium mixture of nickel and Ni_3P.

The changes in resistivity in region (c) display no accompanying change in the magnetization or the diffraction pattern. The phases formed during crystallization are sometimes strained and this step may be associated with internal structural changes due to recrystallization.

ACKNOWLEDGMENTS

The authors are very appreciative of the staff and the Director, Professor Kailash Chandra, USIC, Roorkee University for providing the facilities at the Instrumentation Centre.

REFERENCES

1 I. Bakonyi, A. Cziraki, I. Nagy and M. Hosso, *Z. Metallkd.*, *77* (1986) 425.
2 I. Bakonyi, L. K. Varga, A. Lovas, E. Toth-Kadar and A. Solyom, *J. Magn. Mater.*, *50* (1985) 111.
3 A. Cziraki, B. Fogarassy, I. Bakonyi, K. Tompa, T. Bagi and Z. Hegediis, *J. Phys. (Paris)*, *41* (1980) C8141.
4 E. Vafuei-Makhsoos, E. L. Thomas and L. E. Toth, *Metall. Trans. A*, *9* (1978) 1449.
5 H. Maeda, *Phys. Soc. Jpn.*, *29* (1970) 311.
6 P. A. Albert, Z. Kovac, H. R. Lilienthal, T. R. Miguire and Y. Nakamura, *J. Appl. Phys.*, *38* (1967) 1258.
7 S. V. S. Tyagi, *Ph.D. Thesis*, Roorkee University, 1987.
8 S. V. S. Tyagi, V. K. Tondon and S. Ray, *Bull. Mater. Sci.*, *8* (3) (1986) 433.
9 S. V. S. Tyagi, V. K. Tondon and S. Ray, *Z. Metallkd.*, *76* (7) (1985) 492.
10 B. G. Bagley and D. Turnbull, *Acta Metall.*, *18* (1970) 857.
11 U. Koster and V. Herold, in H. J. Guntherodt and H. Beck (eds.), *Topics in Applied Physics*, Vol. 46, *Glassy Metals I*, Springer, Berlin, 1981.